

EXPERIMENT NO. 13

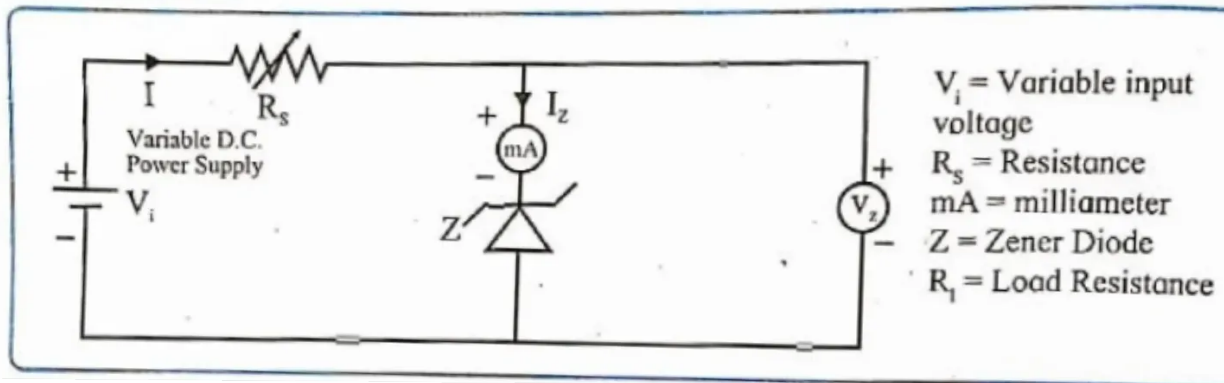
CHARACTERISTICS OF ZENER DIODE

Aim: To study the characteristics of Zener diode in reverse bias and determine its reverse breakdown voltage.

Apparatus: DC power supply, Milliammeter, Voltmeter, Connecting wires etc

Theory: The Zener diode is used to regulate the voltage across a load when there are variations in the supply or load current. The Zener regulator is connected between the output of the bridge rectifier with the filter & the load. The Zener regulator consists of a current limiting resistor R_s connected in series with the input voltage V_s & the Zener diode is connected in parallel with the load R_L . The Zener diode is reverse biased. It is selected with a breakdown voltage V_z equal to the desired across the load. $R_s = V_s - V_z$ As long as V_s is greater then V_z is the Zener diode operates in the breakdown region & output voltage remains constant.

Circuit diagram:



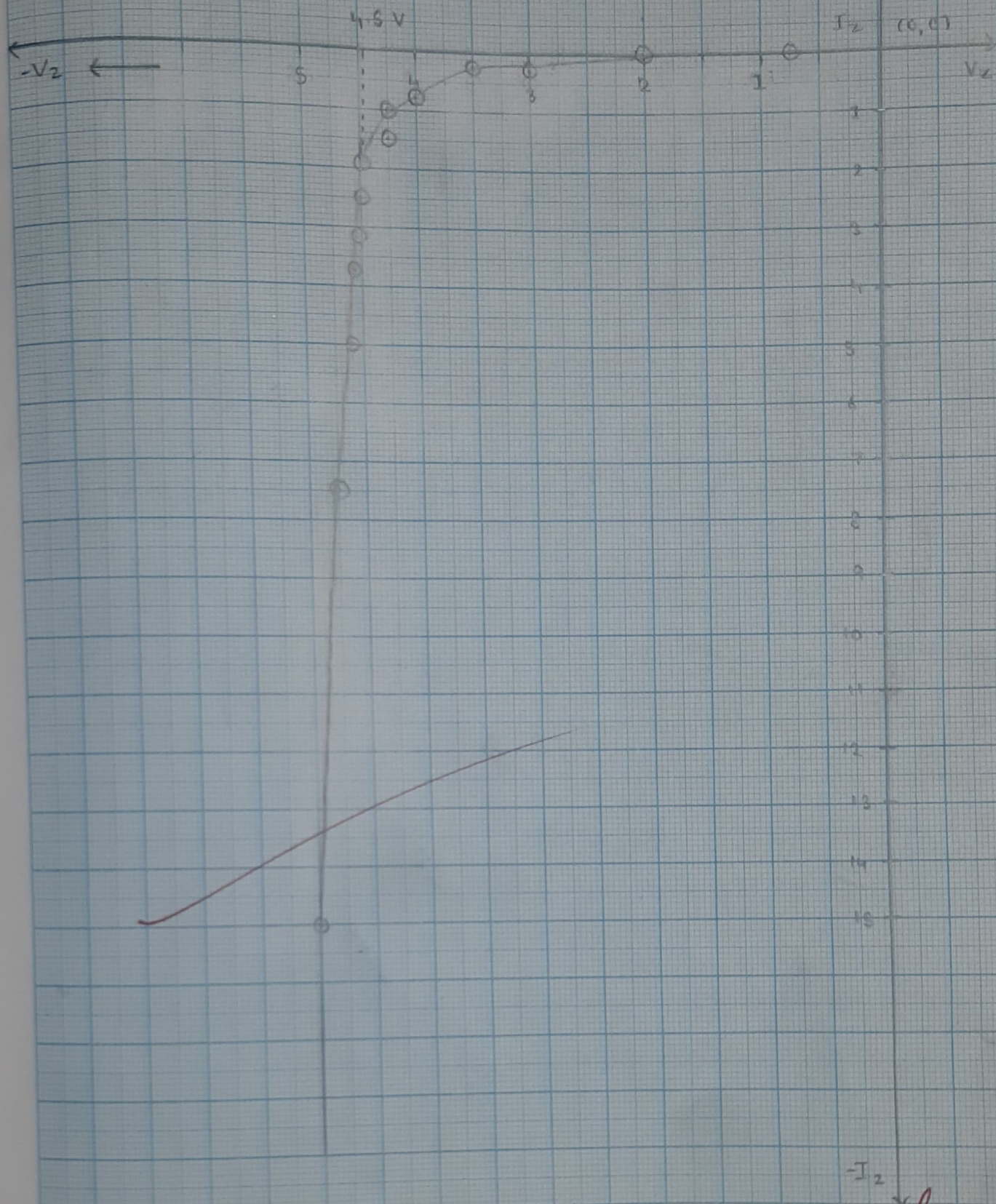
Procedure:

1. Connect the circuit as shown in the circuit diagram.
2. Vary the voltage (V_i) of the D.C. supply as shown in the observation table.
3. Note the current through the Zener diode (I_z).
4. Measure the voltage across Zener diode (V_z)
5. Plot a graph of I_z (-ve Y axis) V_z V (-ve X axis)
6. Note the breakdown voltage (V_B).

Observation table:

Sr. No.	INPUT VOLTAGE (V_i) volts	CURRENT THROUGH ZENER I_z (mA)	OUTPUT VOLTAGE ACROSS ZENER (V_z) volts
1	1 v	0 mA	0 v
2	2 v	0 mA	0.75 v
3	3 v	0 mA	2 v
4	4 v	0.25 mA	3 v
5	5 v	0.25 mA	3.5 v
6	6 v	0.75 mA	4 v
7	7 v	1 mA	4.25 v
8	8 v	1.5 mA	4.25 v
9	9 v	2 mA	4.5 v
10	10 v	2.5 mA	4.5 v

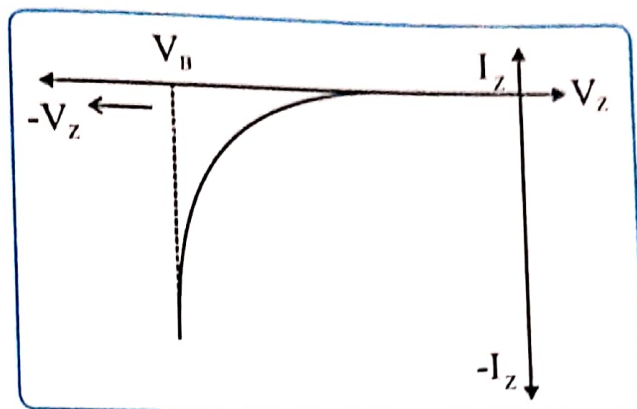
Scale :
 X-axis : 1 unit = 2 mA
 Y-axis : 2 unit = 1 V



Sign of teacher

11	11	3.25 mA	4.5 V
12	12	3.75 mA	4.5 V
13	13	5.0 mA	4.5 V
14	14	7.5 mA	4.75 V
15	15	15 mA	5 V

Graph:



Result: Breakdown voltage of zener diode (V_B) = 4.5 volt

Precautions:

1. All the connections should be tight.
2. Input voltage should be adjusted in suitable range.
3. Increase reverse voltage gradually.

Questions

1. How is the breakdown voltage related to doping level of a diode?

The breakdown voltage of a diode is inversely proportional to its doping level. As the doping concentration increases, the depletion layer becomes thinner and the breakdown voltage decreases.

2. State the applications of Zener diode.

A Zener diode is used as a voltage regulator to maintain a constant output voltage. It is also used for overvoltage protection, in clipping and clamping circuits, and as a reference voltage source in electronic circuits.

3. Distinguish between zener diode and rectifier diode.

A Zener diode is specially designed to operate in the reverse breakdown region and is mainly used for voltage regulation. It is heavily doped and has a low breakdown voltage. In contrast, a rectifier diode is designed to conduct in the forward bias direction and is used for converting AC into DC. It is lightly doped, has a higher breakdown voltage, and may get damaged if operated in the reverse breakdown region.

Remark and sign of teacher :

Excellent
25/1/23